

## WUGANG, China

WMO: 578530

Lat: 26.7383N

Lon: 110.6370E

Elev: 1115

StdP: 14.11

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 29.6       | 32.0 | 18.5                          | 14.6 | 40.9 | 22.7 | 17.8 | 39.4 | 10.7                  | 44.9 | 9.3 | 44.7 | 2.9                   | 0    | 0.321 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.2                   | 95.0            | 76.7 | 93.1 | 76.4 | 91.1 | 76.0 | 79.2                | 88.8 | 78.4 | 87.7 | 77.8 | 86.6 | 5.5                  | 220  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 77.0                            | 146.6 | 82.7 | 76.3 | 142.9 | 82.2 | 75.5 | 139.4 | 81.7 | 43.9          | 88.8 | 43.0 | 88.3 | 42.3 | 86.9 | 83.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 10.3              | 8.8  | 7.5 | 27.5                       | 98.0 | 2.6                | 1.8 | 25.7                                               | 99.3 | 24.2       | 100.3 | 22.8       | 101.3 | 20.9       | 102.6 |
|                   |      |     | 26.3                       | 80.7 | 2.6                | 1.2 | 24.4                                               | 81.6 | 22.9       | 82.3  | 21.4       | 83.0  | 19.5       | 83.9  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 64.1  | 42.6  | 46.7  | 54.1  | 64.9  | 72.4  | 78.0  | 82.7  | 81.7  | 75.4  | 66.2  | 56.5  | 46.7 | (6)  |
| (7)  |                                                                   | DBStd       | 15.18 | 6.81  | 8.38  | 8.11  | 7.84  | 5.63  | 4.70  | 4.00  | 4.40  | 6.18  | 6.22  | 6.82  | 5.73 | (7)  |
| (8)  |                                                                   | HDD50       | 600   | 246   | 151   | 48    | 3     | 0     | 0     | 0     | 0     | 0     | 19    | 133   |      | (8)  |
| (9)  |                                                                   | HDD65       | 2559  | 694   | 515   | 348   | 98    | 7     | 0     | 0     | 4     | 59    | 265   | 569   |      | (9)  |
| (10) |                                                                   | CDD50       | 5741  | 17    | 59    | 176   | 449   | 694   | 840   | 1014  | 983   | 762   | 503   | 214   | 30   | (10) |
| (11) |                                                                   | CDD65       | 2224  | 0     | 2     | 12    | 94    | 236   | 390   | 549   | 518   | 316   | 97    | 10    | 0    | (11) |
| (12) |                                                                   | CDH74       | 20403 | 1     | 19    | 79    | 619   | 1641  | 3311  | 6238  | 5400  | 2542  | 494   | 59    | 0    | (12) |
| (13) |                                                                   | CDH80       | 7851  | 0     | 3     | 18    | 168   | 531   | 1111  | 2754  | 2251  | 902   | 105   | 8     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 3.1   | 2.8   | 3.1   | 3.1   | 3.5   | 3.3   | 3.2   | 3.6   | 3.5   | 3.1   | 2.8   | 2.7   | 2.8  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 54.3  | 2.4   | 2.9   | 4.1   | 6.8   | 8.1   | 7.7   | 4.4   | 5.4   | 3.5   | 4.1   | 3.3   | 1.8  | (15) |
| (16) |                                                                   | PrecMax     | 76.5  | 5.2   | 6.8   | 8.0   | 14.1  | 15.1  | 15.5  | 9.3   | 12.0  | 10.9  | 11.8  | 10.8  | 3.6  | (16) |
| (17) |                                                                   | PrecMin     | 42.0  | 0.2   | 0.4   | 1.4   | 2.2   | 2.5   | 2.4   | 1.3   | 0.4   | 0.1   | 0.2   | 0.1   | 0.0  | (17) |
| (18) |                                                                   | PrecStd     | 7.5   | 1.2   | 1.5   | 1.5   | 2.7   | 3.4   | 3.1   | 1.8   | 2.6   | 2.4   | 2.6   | 2.3   | 1.1  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 66.3  | 76.8  | 81.6  | 89.1  | 92.2  | 93.6  | 97.4  | 97.3  | 94.6  | 87.4  | 80.8  | 68.0 | (19) |
| (20) |                                                                   | MCWB        | 55.3  | 61.6  | 66.1  | 68.9  | 72.9  | 76.8  | 77.2  | 76.6  | 75.5  | 70.0  | 66.8  | 55.2  |      | (20) |
| (21) |                                                                   | 2%          | DB    | 60.9  | 68.9  | 75.3  | 84.7  | 89.1  | 91.1  | 95.1  | 95.0  | 91.5  | 82.8  | 74.7  | 63.4 | (21) |
| (22) |                                                                   | MCWB        | 51.7  | 57.9  | 63.6  | 69.4  | 71.6  | 76.6  | 77.2  | 76.5  | 75.1  | 68.5  | 61.6  | 52.6  |      | (22) |
| (23) |                                                                   | 5%          | DB    | 56.5  | 63.4  | 70.5  | 81.0  | 85.9  | 88.8  | 93.3  | 92.7  | 88.5  | 79.2  | 70.8  | 59.4 | (23) |
| (24) |                                                                   | MCWB        | 49.2  | 55.2  | 60.5  | 68.1  | 71.3  | 76.1  | 77.0  | 76.5  | 74.1  | 66.5  | 60.4  | 50.7  |      | (24) |
| (25) |                                                                   | 10%         | DB    | 52.7  | 58.4  | 66.0  | 77.1  | 82.7  | 86.3  | 91.3  | 90.3  | 85.3  | 76.1  | 67.3  | 55.6 | (25) |
| (26) |                                                                   | MCWB        | 47.2  | 52.4  | 58.4  | 66.9  | 70.9  | 75.5  | 76.6  | 76.3  | 73.1  | 65.3  | 58.7  | 48.8  |      | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 58.4  | 62.7  | 68.9  | 74.2  | 77.9  | 79.6  | 80.4  | 80.1  | 78.3  | 74.6  | 68.8  | 59.5 | (27) |
| (28) |                                                                   | MCD B       | 64.3  | 74.8  | 79.2  | 83.2  | 85.4  | 88.6  | 90.3  | 90.1  | 89.1  | 82.6  | 78.1  | 62.9  |      | (28) |
| (29) |                                                                   | 2%          | WB    | 53.6  | 59.5  | 65.4  | 72.2  | 76.1  | 78.5  | 79.3  | 78.9  | 77.2  | 71.8  | 65.2  | 54.9 | (29) |
| (30) |                                                                   | MCD B       | 58.6  | 67.6  | 73.1  | 80.6  | 83.7  | 86.6  | 89.1  | 88.8  | 87.1  | 77.6  | 70.9  | 59.7  |      | (30) |
| (31) |                                                                   | 5%          | WB    | 50.4  | 56.2  | 62.6  | 70.3  | 74.6  | 77.7  | 78.6  | 78.1  | 76.0  | 69.7  | 62.8  | 51.8 | (31) |
| (32) |                                                                   | MCD B       | 54.8  | 61.7  | 67.7  | 78.1  | 81.4  | 85.1  | 88.2  | 87.5  | 84.8  | 75.5  | 67.7  | 57.6  |      | (32) |
| (33) |                                                                   | 10%         | WB    | 47.7  | 53.2  | 59.5  | 68.5  | 73.1  | 76.9  | 77.9  | 77.4  | 74.9  | 67.8  | 60.2  | 49.4 | (33) |
| (34) |                                                                   | MCD B       | 52.1  | 57.6  | 64.4  | 75.3  | 79.5  | 83.7  | 86.9  | 86.1  | 82.6  | 73.6  | 65.2  | 54.2  |      | (34) |
| (35) | Mean Daily Temperature Range                                      |             | MDBR  | 8.9   | 9.9   | 10.9  | 12.7  | 12.6  | 11.2  | 13.2  | 13.1  | 12.9  | 11.9  | 12.1  | 10.9 | (35) |
| (36) |                                                                   | 5% DB       | MCDBR | 19.1  | 20.8  | 21.2  | 20.4  | 19.6  | 15.7  | 17.4  | 17.6  | 18.3  | 18.3  | 19.8  | 21.0 | (36) |
| (37) |                                                                   | MCWBR       | 10.4  | 10.8  | 9.7   | 7.9   | 6.8   | 5.0   | 4.7   | 4.5   | 5.5   | 6.5   | 8.7   | 10.6  | (37) |      |
| (38) |                                                                   | 5% WB       | MCDBR | 16.0  | 17.9  | 17.7  | 17.1  | 14.1  | 12.8  | 14.3  | 14.4  | 15.2  | 13.5  | 15.0  | 16.7 | (38) |
| (39) |                                                                   | MCWBR       | 9.4   | 10.2  | 9.7   | 8.3   | 6.4   | 4.8   | 4.8   | 4.6   | 5.2   | 6.6   | 7.9   | 9.5   | (39) |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.665 | 0.691 | 0.766 | 0.734 | 0.637 | 0.615 | 0.557 | 0.639 | 0.674 | 0.731 | 0.652 | 0.646 | (40) |      |
| (41) |                                                                   | taud        | 1.522 | 1.519 | 1.436 | 1.538 | 1.764 | 1.833 | 1.996 | 1.760 | 1.659 | 1.499 | 1.607 | 1.574 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 182   | 193   | 189   | 202   | 223   | 227   | 240   | 220   | 206   | 181   | 184   | 179   | (42) |      |
| (43) |                                                                   | Edh at Noon | 80    | 86    | 98    | 91    | 72    | 67    | 57    | 72    | 77    | 86    | 73    | 73    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 506   | 614   | 785   | 1002  | 1134  | 1167  | 1541  | 1387  | 1150  | 914   | 755   | 601   | (44) |      |
| (45) |                                                                   | RadStd      | 121   | 158   | 109   | 149   | 192   | 167   | 168   | 114   | 124   | 124   | 129   | 142   | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only           | +0.97 | N/A     | -2.70  | +1.28   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (7 neighbors) | +0.52 | N/A     | N/A    | +1.06   | N/A   | N/A   | N/A         | N/A   | +212  | +128  |

Nomenclature: See separate page